

$$y \sim N(\mu, \sigma^2)$$

$\uparrow \quad \uparrow$
 unknown

Noninformative Prior $\rightarrow$ Jeffreys's Invariance Principle

$p(\theta) \propto [J(\theta)]^{1/2}$ where $J(\theta) = -E\left(\frac{d^2 \log p(y|\theta)}{d\theta^2} \mid \theta\right)$

$$\theta = (\mu, \sigma)$$

$$L := p(y|\theta) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{1}{2\sigma^2}(y-\mu)^2\right)$$

$$\log(L) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2}(y-\mu)^2$$

$$= -\frac{1}{2} \log(2\pi) - \log(\sigma) - \frac{1}{2\sigma^2}(y-\mu)^2$$

$$\frac{\partial \log(L)}{\partial \mu} = \frac{1}{\sigma^2}(y-\mu) \quad \frac{\partial \log(L)}{\partial \sigma} = -\frac{1}{\sigma} + \frac{1}{\sigma^3}(y-\mu)^2$$

$$\frac{\partial^2 \log(L)}{(\partial \mu)^2} = -\frac{1}{\sigma^2} \quad \frac{\partial^2 \log(L)}{\partial \mu \partial \sigma} = -\frac{2(y-\mu)}{\sigma^3} \quad \frac{\partial^2 \log(L)}{(\partial \sigma)^2} = \frac{1}{\sigma^2} - \frac{3}{\sigma^4}(y-\mu)^2$$

$$J(\theta) = \left(-E \left[\det \begin{vmatrix} -\frac{1}{\sigma^2} & \frac{-2(y-\mu)}{\sigma^3} \\ \frac{-2(y-\mu)}{\sigma^3} & \frac{1}{\sigma^2} - \frac{3}{\sigma^4}(y-\mu)^2 \end{vmatrix} \right] \right)^{1/2}$$

$$= \left(-E \left[-\frac{1}{\sigma^4} + \frac{3}{\sigma^6}(y-\mu)^2 - \frac{4(y-\mu)^2}{\sigma^6} \right] \right)^{1/2}$$

$$= \left(\frac{1}{\sigma^4} + \frac{1}{\sigma^4} \right)^{1/2} = \left(\frac{2}{\sigma^4} \right)^{1/2} = \frac{\sqrt{2}}{\sigma^2} \propto \frac{1}{\sigma^2}$$

$$E(y-\mu)^2 = \sigma^2$$

$$p(\theta) \propto \frac{1}{\sigma^2}$$

Noninformative prior.

Posterior:

$$p(\theta|y) \propto p(y|\theta) p(\theta) \quad \theta = (\mu, \sigma^2)$$

$$\propto (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2\right) \cdot \frac{1}{\sigma^2}$$

$$\propto (\sigma^2)^{-n/2-1} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2\right)$$

$$\begin{aligned} \sum_{i=1}^n (y_i - \mu)^2 &= \sum_{i=1}^n ((y_i - \bar{y}) + (\bar{y} - \mu))^2 \\ &= \sum_{i=1}^n \left[(y_i - \bar{y})^2 + 2(y_i - \bar{y})(\bar{y} - \mu) + (\bar{y} - \mu)^2 \right] \\ &= \sum_{i=1}^n (y_i - \bar{y})^2 + 2(\bar{y} - \mu) \sum_{i=1}^n (y_i - \bar{y}) + n(\bar{y} - \mu)^2 \end{aligned}$$

$$p(\mu, \sigma^2|y) \propto (\sigma^2)^{-n/2-1} \exp\left(-\frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2 \right)\right)$$

Sample variance: $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$

$$p(\mu, \sigma^2|y) \propto (\sigma^2)^{-n/2-1} \exp\left(-\frac{1}{2\sigma^2} \left(s^2(n-1) + n(\bar{y} - \mu)^2 \right)\right)$$

The parameter we "care" about is μ .

$$p(\mu | \sigma^2, y) \propto \exp\left(-\frac{n}{2\sigma^2} (\mu - \bar{y})^2\right)$$

Joint posterior distribution.

Normalized:

$$p(\mu | \sigma^2, y) = \sqrt{\frac{n}{2\pi\sigma^2}} \exp\left(-\frac{n}{2\sigma^2}(\mu - \bar{y})^2\right)$$

$$\mu | \sigma^2, y \sim N(\bar{y}, \sigma^2/n)$$

conditional posterior

Marginal posterior:

$$p(\sigma^2 | y) \propto \int_{-\infty}^{\infty} p(\mu, \sigma^2 | y) d\mu$$

$$\propto \int_{-\infty}^{\infty} \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2}((n-1)s^2 + n(\bar{y} - \mu)^2)\right) d\mu$$

$$\propto \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2}(n-1)s^2\right) \cdot \underbrace{\int_{-\infty}^{\infty} \exp\left(-\frac{n}{2\sigma^2}(\bar{y} - \mu)^2\right) d\mu}_{\sqrt{\frac{2\pi\sigma^2}{n}}}$$

$$p(\sigma^2 | y) \propto \sqrt{\frac{2\pi\sigma^2}{n}} \sigma^{-n-2} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right)$$

$$\propto \sigma^{-n-1} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right) \rightarrow \text{Inverse Gamma Distribution}$$

$$a = \frac{n}{2} - 1 \quad b = \frac{(n-1)s^2}{2}$$

Posterior prediction density

$$p(\tilde{y}|y) = \iint p(\tilde{y}|\mu, \sigma^2) p(\mu, \sigma^2|y) d\mu d\sigma^2$$

$$p(\mu, \sigma^2|y) \propto \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{y}-\mu)^2]\right)$$

likelihood $\tilde{y}|\mu, \sigma^2 \sim N(\mu, \sigma^2)$ so...

$$\begin{aligned} p(\tilde{y}|y) &\propto \int_0^\infty \int_{-\infty}^\infty \sigma^{-1} \exp\left(-\frac{1}{2\sigma^2}(\tilde{y}-\mu)^2\right) \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{y}-\mu)^2]\right) d\mu d\sigma^2 \\ &= \int_0^\infty \sigma^{-n-3} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right) \int_{-\infty}^\infty \exp\left(-\frac{1}{2\sigma^2}((\tilde{y}-\mu)^2 + n(\bar{y}-\mu)^2)\right) d\mu d\sigma^2 \end{aligned}$$

$$(\tilde{y}-\mu)^2 + n(\bar{y}-\mu)^2 = \tilde{y}^2 - 2\tilde{y}\mu + \mu^2 + n\bar{y}^2 - 2n\bar{y}\mu + n\mu^2$$

$$= (n+1)\mu^2 - 2\mu(\tilde{y} + n\bar{y}) + \tilde{y}^2 + n\bar{y}^2$$

$$= (n+1)\mu^2 - 2\mu(\tilde{y} + n\bar{y}) + \frac{(n+1)\tilde{y}^2}{(n+1)} + \frac{(n+1)n\bar{y}^2}{(n+1)}$$

$$= (n+1)\mu^2 - 2\mu(\tilde{y} + n\bar{y}) + \frac{(n+1)\tilde{y}^2 + (n+1)n\bar{y}^2 + 2n\bar{y}\tilde{y} - 2n\bar{y}\tilde{y}}{(n+1)}$$

$$= (n+1)\mu^2 - 2\mu(\tilde{y} + n\bar{y}) + \frac{\tilde{y}^2 + 2n\bar{y}\tilde{y} + n^2\bar{y}^2 + n\tilde{y}^2 - 2n\bar{y}\tilde{y} + n\tilde{y}^2}{(n+1)}$$

$$= (n+1)\mu^2 - 2\mu(\tilde{y} + n\bar{y}) + \frac{(\tilde{y} + n\bar{y})^2 + n(\bar{y} - \tilde{y})^2}{(n+1)}$$

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$$= (n+1) \left[\mu^2 - \frac{2\mu(\tilde{y} + n\bar{y})}{n+1} + \frac{(\tilde{y} + n\bar{y})^2}{(n+1)^2} \right] + \frac{n(\bar{y} - \tilde{y})^2}{n+1}$$

$$= (n+1) \left(\mu - \frac{(\tilde{y} + n\bar{y})}{n+1} \right)^2 + \frac{n(\bar{y} - \tilde{y})^2}{(n+1)}$$

$$= \int_0^\infty \sigma^{-n-3} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right) \int_{-\infty}^\infty \exp\left(-\frac{1}{2\sigma^2} \left((n+1) \left(\mu - \frac{(\tilde{y} + n\bar{y})}{n+1} \right)^2 + \frac{n(\bar{y} - \tilde{y})^2}{(n+1)} \right) \right) d\mu d\sigma^2$$

$$= \int_0^\infty \sigma^{-n-3} \exp\left(-\frac{(n-1)s^2}{2\sigma^2} - \left(\frac{n}{n+1}\right) \frac{(\bar{y} - \tilde{y})^2}{2\sigma^2}\right) \underbrace{\int_{-\infty}^\infty \exp\left(-\frac{n+1}{2\sigma^2} (\mu - \mu)^2\right) d\mu}_{= \sqrt{\frac{2\pi\sigma^2}{n+1}}} d\sigma^2$$

$$= \int_0^\infty \sigma^{-n-2} \cdot \sqrt{\frac{2\pi}{n+1}} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right) \exp\left(-\frac{1}{2\sigma^2} \cdot \frac{n}{n+1} \cdot (\bar{y} - \tilde{y})^2\right) d\sigma^2$$

$$= \sqrt{\frac{2\pi}{n+1}} \int_0^\infty (\sigma^2)^{-n/2-1} \exp\left(-\frac{1}{2\sigma^2} \left[s^2(n-1) + \frac{n}{n+1} (\bar{y} - \tilde{y})^2 \right] \right) d\sigma^2$$

Gamma Distribution: $\int_0^\infty x^{a-1} e^{-bx} dx = \frac{\Gamma(a)}{b^a}$

$$u = \frac{1}{\sigma^2} \Rightarrow \sigma^2 = \frac{1}{u} \Rightarrow d\sigma^2 = -\frac{1}{u^2} du$$

$$= \sqrt{\frac{2\pi}{n+1}} \int_\infty^0 u^{n/2+1} \exp\left(-\frac{1}{2} u \left[s^2(n-1) + \frac{n}{n+1} (\bar{y} - \tilde{y})^2 \right] \right) -\frac{1}{u^2} du$$

$$= -\sqrt{\frac{2\pi}{n+1}} \int_0^\infty u^{n/2-1} \exp\left(-\frac{1}{2}u\left[s^2(n-1) + \frac{n}{n+1}(\bar{y}-\tilde{y})^2\right]\right) du$$

$$\int_0^\infty x^{a-1} e^{-bx} dx = \frac{\Gamma(a)}{b^a}$$

$$a = n/2 \quad b = \frac{1}{2}\left[s^2(n-1) + \frac{n}{n+1}(\bar{y}-\tilde{y})^2\right]$$

$$\tilde{y}|y \propto -\sqrt{\frac{2\pi}{n+1}} \frac{\Gamma(n/2)}{\left(\frac{1}{2}\left[s^2(n-1) + \frac{n}{n+1}(\bar{y}-\tilde{y})^2\right]\right)^{n/2}}$$

From here, you can show that

$$\tilde{y}|y \sim t_{n-1}\left(\bar{y}, \left(\frac{n+1}{n}\right)s^2\right)$$

Reminder of a t -distribution

$$\frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right)\sqrt{(n-1)\pi\sigma^2}} \left(1 + \frac{1}{n-1}\left(\frac{\tilde{y}-\bar{y}}{s\sqrt{\frac{n+1}{n}}}\right)^2\right)^{-n/2}$$

$$\begin{aligned} \tilde{y}|y &\propto -\sqrt{\frac{2\pi}{n+1}} \Gamma\left(\frac{n}{2}\right) \left(\frac{1}{2}\left[s^2(n-1) + \frac{n}{n+1}(\bar{y}-\tilde{y})^2\right]\right)^{-n/2} \\ &= -\sqrt{\frac{2\pi}{n+1}} \Gamma\left(\frac{n}{2}\right) 2^{n/2} s^2(n-1) \left(1 + \frac{1}{n-1}\left(\frac{\bar{y}-\tilde{y}}{s\sqrt{\frac{n+1}{n}}}\right)^2\right)^{-n/2} \end{aligned}$$